

# ODEs

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# ODE Zoology

- ▶ Order
- ▶ Linearity
- ▶ Homogeneity
- ▶ Autonomy
- ▶ Constant/Variable Coefficients
- ▶ Exactness
- ▶ Separability

- ▶ **Definition:** The order of an ordinary differential equation (ODE) is the index of the highest derivative of the unknown function.
- ▶ **First-Order ODE:**
  - ▶ Example:  $y' = f(x, y)$ .
- ▶ **Second-Order ODE:**
  - ▶ Example:  $y'' + p(x)y' + q(x)y = g(x)$ .
- ▶ Higher-order ODEs can be reduced to systems of first-order ODEs.

# Converting nth Order ODE to System of First-Order ODEs

## ► General nth Order ODE:

►  $y^{(n)} = F(x, y, y', y'', \dots, y^{(n-1)})$

► Introduce new variables:

$$y_1 = y, \quad y_2 = y', \dots, y_n = y^{(n-1)}$$

► The system of first-order ODEs:

$$y_1' = y_2,$$

$$y_2' = y_3,$$

$$\vdots$$

$$y_{n-1}' = y_n,$$

$$y_n' = F(x, y_1, y_2, \dots, y_n)$$

$$a_n(x) \frac{d^n y}{dx^n} + a_{n-1}(x) \frac{d^{n-1} y}{dx^{n-1}} + \dots + a_1(x) \frac{dy}{dx} + a_0(x)y = g(x) \quad (1)$$

where  $a_i(x)$  and  $g(x)$  are given functions.

# ODE Zoology. Homogeneity

- ▶ **Definition:** A differential equation is called homogeneous if all its terms are a function of the dependent variable and its derivatives only, and it equals zero.
- ▶ **Linear Homogeneous ODE:**
  - ▶ Example:  $y'' + p(x)y' + q(x)y = 0$ .
- ▶ **Non-linear Homogeneous ODE:**
  - ▶ Example:  $y'' + y^3 = 0$ .
- ▶ **Properties:**
  - ▶ Solutions can be scaled: if  $y(x)$  is a solution, then  $c \cdot y(x)$  is also a solution for any constant  $c$ .

## ODE Zoology. Linear + Homogeneous

$$a_n(x) \frac{d^n y}{dx^n} + a_{n-1}(x) \frac{d^{n-1} y}{dx^{n-1}} + \dots + a_1(x) \frac{dy}{dx} + a_0(x)y = 0 \quad (2)$$

- ▶ **Superposition Principle:** If  $y_1(x)$  and  $y_2(x)$  are solutions, then  $c_1 y_1(x) + c_2 y_2(x)$  is also a solution for any constants  $c_1$  and  $c_2$ .
- ▶ **Existence of a Basis:** The set of all solutions forms a vector space, with a basis consisting of linearly independent solutions.
- ▶ **Wronskian**  $\approx$  Determinant of basis vectors.  
If  $W(y_1, y_2, \dots, y_n)(x) \neq 0$  for some  $x$ , then these solutions are linearly independent.



- ▶ **Definition:** An ODE is autonomous if it does not explicitly depend on the independent variable.
- ▶ **Autonomous Form:**
  - ▶  $\frac{dy}{dx} = f(y)$  (first-order)
  - ▶  $\frac{d^2y}{dx^2} = f(y, y')$  (second-order)
- ▶ **Non-Autonomous Form:**
  - ▶  $\frac{dy}{dx} = f(x, y)$  (explicitly depends on  $x$ )
- ▶ Time-translation invariance: if  $y(x)$  is a solution, then  $y(x + c)$  is also a solution

# ODE Zoology. Autonomy on Dynamical Systems

- ▶ **Autonomous ODEs:** Can "advance by themselves"
  - ▶ Self-contained system, no external driving function needed
- ▶ **Non-Autonomous ODEs:** Require external information
  - ▶ Need explicit form/mapping of how system depends on  $x$
  - ▶ Externally driven or forced systems

## ODE Zoology. Constness of Coefficients

$$a_n y^{(n)} + a_{n-1} y^{(n-1)} + \dots + a_1 y' + a_0 y = g(x) \quad (3)$$

## ► Exact Differential Equation:

- Form:  $M(x, y)dx + N(x, y)dy = 0$
- **Exactness condition:**  $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$
- Represents total differential:  $dF(x, y) = 0$  for some function  $F$

# ODEs. Analytical Methods

- ▶ Separation of Variables
- ▶ Characteristic Equation
- ▶ Variation of Parameters
- ▶ and more...

# Separation of Variables

► **Applies to:** First-order ODEs of the form  $\frac{dy}{dx} = g(x)h(y)$

► **Method:**

1. Rewrite as  $\frac{dy}{h(y)} = g(x)dx$
2. Integrate both sides:  $\int \frac{dy}{h(y)} = \int g(x)dx$

► **Example:**  $\frac{dy}{dx} = xy$

►  $\frac{dy}{y} = xdx$

►  $\ln |y| = \frac{x^2}{2} + C$

►  $y = Ae^{x^2/2}$

# Characteristic Equation Method

- ▶ **General form:**  $a_n y^{(n)} + a_{n-1} y^{(n-1)} + \dots + a_1 y' + a_0 y = 0$
- ▶ **Characteristic Eq:**  $a_n z^n + a_{n-1} z^{n-1} + \dots + a_1 z + a_0 = 0$
- ▶ **General Solution cases:**

- ▶ **Distinct Roots** ( $z_1 \neq z_2 \neq \dots \neq z_n$ ):

$$y(x) = C_1 e^{z_1 x} + C_2 e^{z_2 x} + \dots + C_n e^{z_n x}$$

- ▶ **Complex Roots** ( $z_j = \alpha + i\beta$ ):

$$y_j(x) = e^{\alpha x} \cdot (C_1 \cos(\beta x) + C_2 \sin(\beta x))$$

- ▶ **Repeated Roots** ( $z_j$ , multiplicity =  $m_j$ ):

$$y_j(x) = (C_1 + C_2 x + \dots + C_{m_j} x^{m_j-1}) e^{z_j x}$$

- ▶ **General Solution:**

$$y(x) = \sum_{j=1}^k (C_{j1} + C_{j2} x + \dots + C_{jm_j} x^{m_j-1}) e^{z_j x}$$

where  $m_j$  is the multiplicity of the  $j$ -th root

# Characteristic Equation Method

## ► General Solution:

$$y(x) = \sum_{j=1}^k (C_{j1} + C_{j2}x + \dots + C_{jm_j}x^{m_j-1}) e^{z_j x}$$

where  $m_j$  is the multiplicity of the  $j$ -th root



# The form of solutions to Linear Non-Homogeneous ODEs

$$L[y] = f(x)$$

where  $L$  is a linear differential operator

► **General Solution Structure:**

$$y = y_h + y_p$$

- $y_h$ : General solution to homogeneous equation  $L[y] = 0$
- $y_p$ : Any particular solution to  $L[y] = f(x)$

► **Why:** If  $y_1, y_2$  are solutions to  $L[y] = f(x)$ , then:

$$L[y_1 - y_2] = L[y_1] - L[y_2] = f(x) - f(x) = 0$$

So  $(y_1 - y_2)$  satisfies the homogeneous equation

So  $y_1 = y_h + y_2$

# Variation of Parameters

## ► General Higher Order Linear ODE:

$$y^{(n)}(x) + \sum_{i=0}^{n-1} a_i(x)y^{(i)}(x) = b(x)$$

## ► Homogeneous Solutions: $y_1(x), y_2(x), \dots, y_n(x)$

## ► Particular Solution: $y_p(x) = \sum_{i=1}^n c_i(x)y_i(x)$

## ► Extra Conditions: $\sum_{i=1}^n c'_i(x)y_i^{(j)}(x) = 0$ for $j = 0, \dots, n-2$

## ► System (Cramer's Rule):

$$\begin{bmatrix} y_1 & \cdots & y_n \\ y_1' & \cdots & y_n' \\ \vdots & \ddots & \vdots \\ y_1^{(n-1)} & \cdots & y_n^{(n-1)} \end{bmatrix} \begin{bmatrix} c_1' \\ c_2' \\ \vdots \\ c_n' \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ b(x) \end{bmatrix}$$

## ► Solution: $c'_i = \frac{W_i(x)}{W(x)}$ where $W$ is the Wronskian

# Variation of Parameters

## ► Linear System:

$$\begin{bmatrix} y_{1H} & \cdots & y_{nH} \\ y'_{1H} & \cdots & y'_{nH} \\ \vdots & \ddots & \vdots \\ y_{1H}^{(n-1)} & \cdots & y_{nH}^{(n-1)} \end{bmatrix} \begin{bmatrix} c'_1 \\ c'_2 \\ \vdots \\ c'_n \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ b(x) \end{bmatrix}$$

- **Cramer's Rule Solution:**  $c'_i = \frac{W_i(x)}{W(x)}$  for  $i = 1, \dots, n$  where
  - $W$ : Wronskian determinant of the basis  $y_{1H} \dots y_{nH}$
  - $W_i$ : Wronskian det. with  $i$ -th column as  $[0, 0, \dots, 0, b(x)]^T$
- **Particular Solution:**  $y_p(x) = \sum_{i=1}^n y_{iH}(x) \int c'_i(x) dx$
- **Complete Solution:**  $y = y_H + y_p$

haha computer go brrr